

Foundation · For every age group

3D Modelling & 3D Printing

From an idea in your head to an object in your hand

Everyone

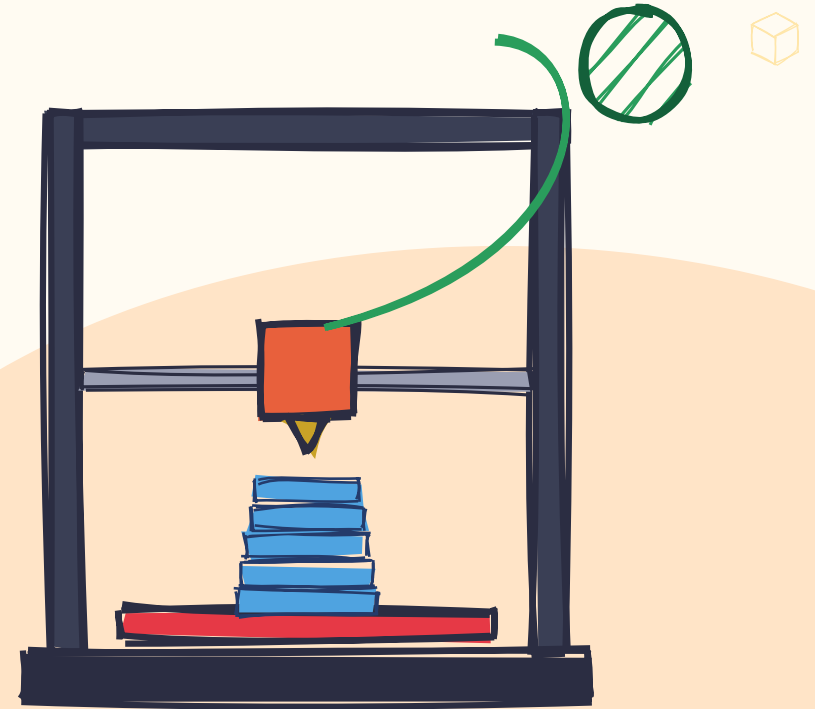
Ages 8+

Ages 12+

Ages 15+

How to use this deck

Coloured badges show the level of each slide. Teach 'Everyone' slides to all, then add the levels that fit your class.



Let's wonder...

What if you could **MAKE** anything?

A design superpower

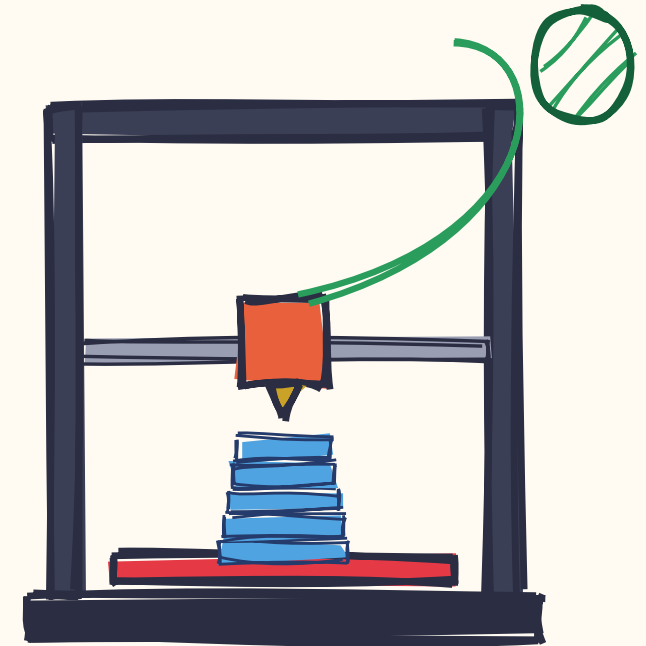
A broken toy part, a phone stand, a robot arm, a name badge.
Today, engineers draw these on a computer and a machine builds them **layer by layer** while they watch.

Buy it?

Carve it from wood?

Design and print it!

Today: you learn how 3D printing works, how to design your own objects, and how to print them **safely and well**.





Learning aims

Today I will learn to...



1

Explain how a 3D printer builds objects **layer by layer**

2

Design a model in **Tinkercad** with shapes and holes

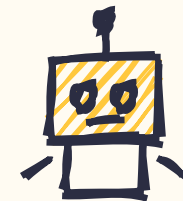
3

Measure in **millimetres** and design parts that **fit**

4

**Slice** a model and choose good print settings

5

Print **safely**, fix common problems and design responsibly**Keywords**3D model · CAD · STL · slicer · layer ·
infill · support · tolerance

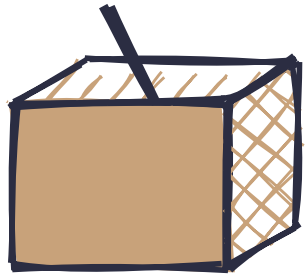


Big idea

Adding, not cutting away

Everyone

Subtractive (old way)

**carve away**

Start with a block, cut away what you don't need. Lots of **waste**.

Additive (3D printing)

**add layers**

Build up thin layers, only where needed. Very **little waste**.

Like stacking pancakes

Each layer is about **0.2 mm** thick, thinner than 2 sheets of paper. A 2 cm tall object has about **100 layers**!



Try it

Stack coins of shrinking size to build a mountain. That is how a printer thinks!





Real world

What do people 3D print?



Prosthetic hands

Volunteers print low-cost hands for children around the world



Rockets & planes

Engine parts too complex to make any other way



Medicine

Dental braces, hearing aids and models of organs for surgeons



Prototypes

Engineers test a design in hours instead of weeks



If you had a 3D printer at home, what is the FIRST useful thing you would print?

Even houses!

Giant printers can print the walls of a house from concrete in a few days.

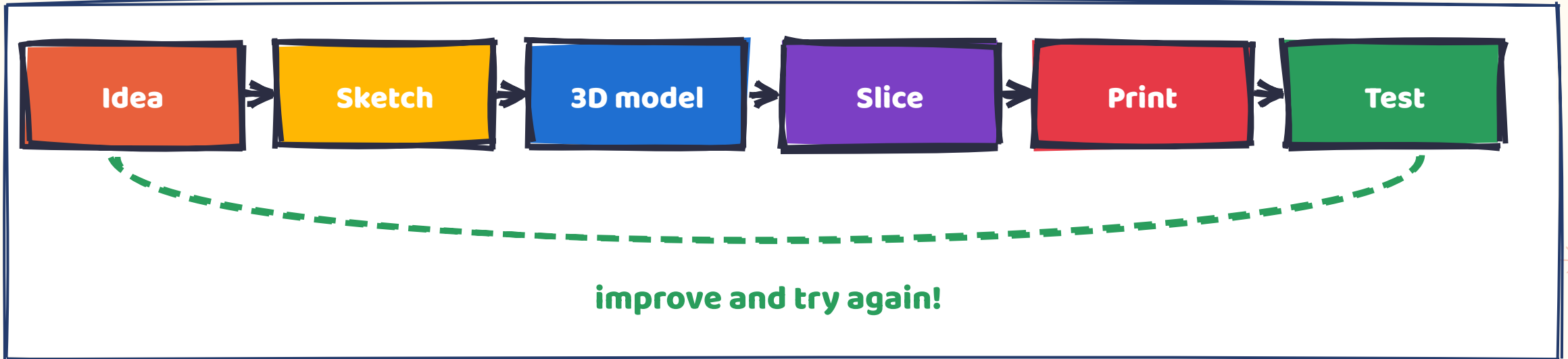




The journey

Everyone

From idea to object in 6 steps



Step	Tool	Typical time
Sketch	paper and pencil	2 minutes
3D model	Tinkercad (CAD software)	20 to 40 minutes
Slice	UltiMaker Cura makes printer instructions (G-code)	2 minutes
Print	the 3D printer	30 minutes to many hours

Patience!

Printing is slow. Plan so prints run overnight or between classes.

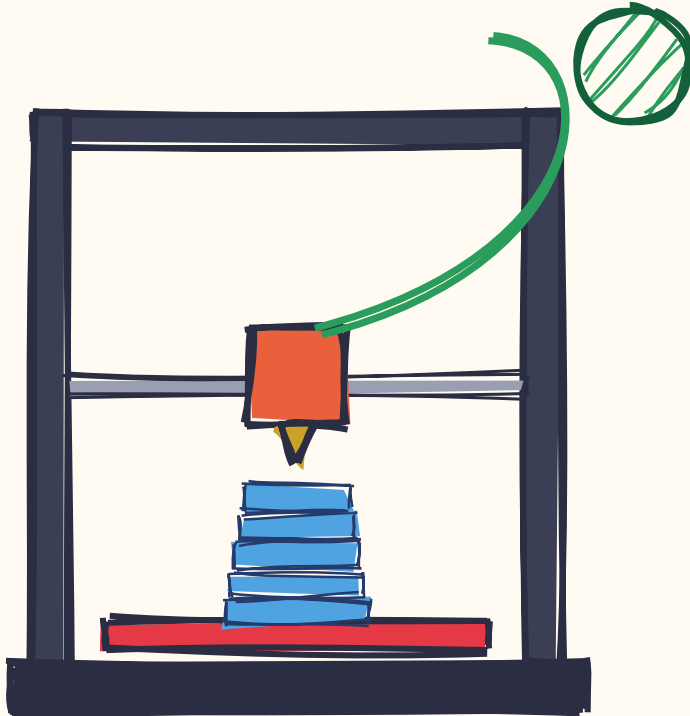


Look inside

Everyone



How an FDM printer works



Part	Job
Filament	plastic string on a spool (1.75 mm thick)
Extruder	motor that pushes filament into the hot end
Nozzle	melts plastic at about 200 °C, hole 0.4 mm wide
Bed	flat plate the print sticks to, often heated to 60 °C
Motors	move in X, Y and Z very precisely

FDM

Fused Deposition Modelling:
melted plastic laid down
(deposited) in lines.

Like a hot glue gun on a robot

The nozzle draws one layer, the bed moves down a tiny bit, and it draws the next layer on top.





Materials

Which plastic should we use?

Filament	What it's like	Nozzle temp	Use for
PLA	easy, low smell, made from plants like corn	about 200 °C	our choice
PETG	tougher, a bit bendy, water-resistant	about 235 °C	outdoor parts
TPU	rubbery and flexible	about 220 °C	phone cases, tyres
ABS	strong but smelly, warps easily	about 240 °C	experts, with ventilation

Cost check

1 kg roll of PLA ≈ Rs 4,600
= about Rs 4.6 per gram.

Keep it dry

Store filament in a bag with silica gel. Damp filament prints badly.





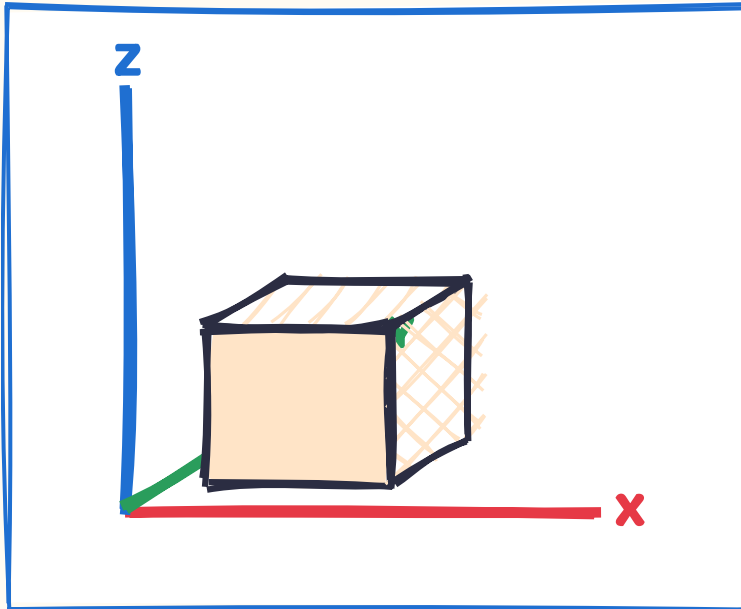
Thinking in 3D

Everyone

Ages 8+



X, Y and Z, measured in millimetres

**X = width**

left and right

Y = depth

front and back

Z = height

up and down

Millimetres (mm)

3D design always uses **mm**. 10 mm = 1 cm. A coin is about 2 mm thick; a bank card is about 54 mm × 86 mm.

Measure tools

Ruler for big parts. A digital caliper measures to 0.1 mm!

Younger groups: just use “wide, deep, tall”. **Older groups:** practise reading X, Y, Z coordinates in Tinkercad.





Our tool

Everyone

Tinkercad: free 3D design in the browser



Drag shapes

box, cylinder, sphere, text



Move

drag, or arrow keys



Rotate

curved arrows on the shape



Resize

white corner handles; type exact mm



Look around

view cube top-left, right-drag to orbit



Undo

Ctrl + Z is your best friend!



Accounts for students

The teacher creates a **Tinkercad Classroom**. Students join with a **class code** and a **nickname**, so under-13s do not need their own email.

Website

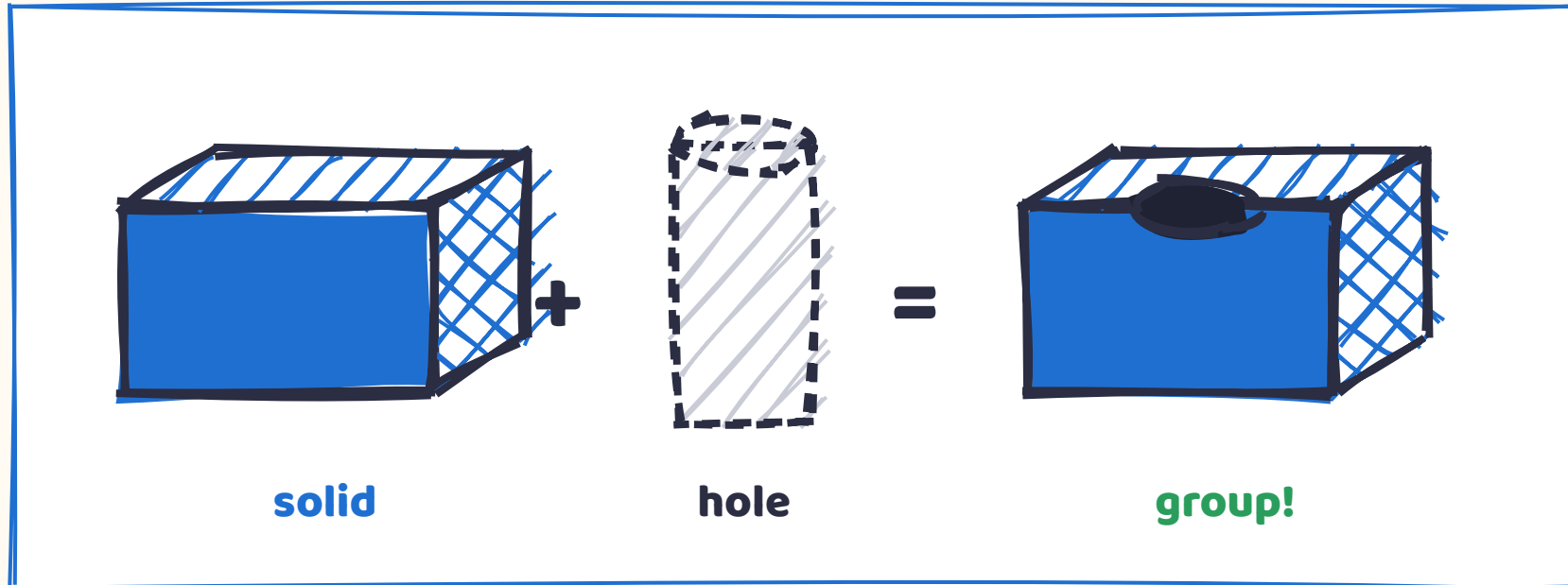
tinkercad.com · works on school laptops, no install needed.





The magic trick

Solids + holes = any shape



Think in pieces

A cup = cylinder minus a smaller cylinder. A keyring = flat shape minus a small hole.

How to do it

1. Place a shape. 2. Place another shape and set it to **Hole**. 3. Select both and click **Group** (Ctrl + G). The hole is cut out!

Ungroup

Made a mistake? Select and Ungroup (Ctrl + Shift + G).





Precision

Align, ruler and exact sizes

Ages 8+

Ages 12+

Align tool (L key)

Select shapes, press **L**, click a dot to line up their centres or edges perfectly.

Ruler (R key)

Drop the ruler on the workplane to type **exact distances** and positions in mm.

Type sizes

Click a white handle and **type** the number, e.g. 40 mm, instead of dragging.

Snap grid

Bottom-right setting. Set it to **1 mm** (or 0.5 mm) for neat designs.

Duplicate & repeat

Ctrl + D duplicates AND repeats the last move.
Great for rows of holes!



Design for printing

Rules that save failed prints

Ages 8+

Ages 12+



Flat base: the bottom must sit flat on the bed.



Overhangs: slopes steeper than 45° need supports underneath.



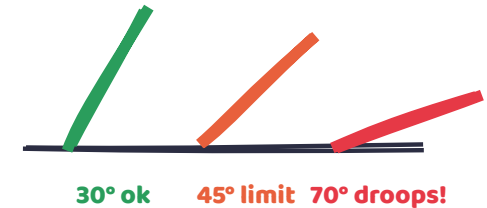
Thick walls: at least 1.2 mm (3 nozzle widths), or they break.



Text: raise or cut letters at least 1 mm and make them big.



Tiny parts: anything under 2 mm is fragile. Keep it chunky!



Rotate it!

Often the best fix is to lay the model on a different side.



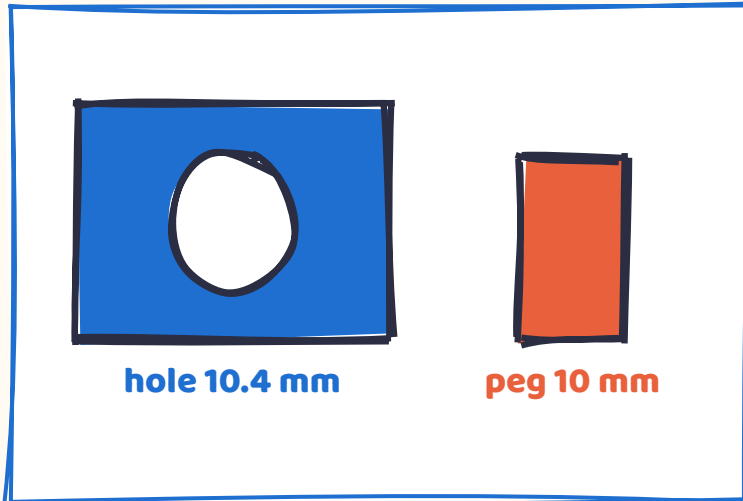


Make parts fit

Tolerance: leave a little gap

Ages 12+

Ages 15+



The problem

A 10 mm peg will **NOT** fit a 10 mm hole.
Melted plastic spreads a little, so holes print **smaller** and pegs print **bigger**.

Test coupon

Before printing a big part, print a small strip with holes of 10.0, 10.2, 10.4 and 10.6 mm.
See which fits best on **your** printer.

Fit	Make hole bigger by
Loose / sliding	+0.4 to 0.5 mm
Snug push fit	+0.2 to 0.3 mm
Tight press fit	+0.1 mm



Measure twice

Use a caliper on the **REAL** part (sensor, motor, screw) before designing.





Ages 8+

Ages 12+



Prepare to print

Export STL, then slice in Cura



1. Export

In Tinkercad: **Export** → **.STL**. An STL describes the surface of your model as thousands of tiny triangles.

2. Slice

Open the STL in **UltiMaker Cura**. Pick the printer and PLA, then click **Slice**.

3. G-code

Cura writes **G-code**: thousands of move and heat commands. Save it to the SD card or USB.



layer 10 of 10

Preview!

Use Cura's **Preview** slider to watch every layer before you print. Spot problems early, for free.

Cura shows

print time and grams of filament. Use them on the cost slide!





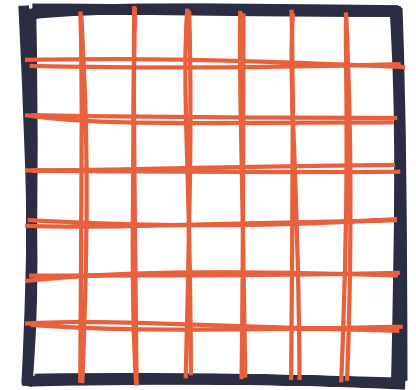
Slicer settings

The 5 settings that matter most

Ages 12+

Ages 15+

Setting	Start with	What it does
Layer height	0.2 mm	smaller = smoother but slower
Infill	15 to 20 %	the inside pattern; more = stronger, slower, heavier
Walls	3 lines (1.2 mm)	outer shell strength
Supports	only if needed	props under overhangs; snap off after
Adhesion	skirt or brim	brim = wide edge that stops corners lifting



grid infill (top view)

100 % infill?

Almost never! It wastes plastic and time for little extra strength.





Maths moment

Ages 8+

Ages 12+



How long, and how much?



Object	Filament	Print time	Cost (Rs 4.6 / g)
Name badge	8 g	35 min	$8 \times 4.6 = \text{Rs } 37$
Sensor mount	15 g	1 h 10 min	$15 \times 4.6 = \text{Rs } 69$
Plant monitor case	45 g	3 h 30 min	$45 \times 4.6 = ?$
Class of 30 badges	$30 \times 8 \text{ g}$	?	?

Plan the queue

One printer, 30 students, 35 minutes each = **17.5 hours** of printing! Print several small items **together** on one bed.

Costs use our lab PLA price (about Rs 4,600 per 1 kg roll). Times and grams are examples: always read the real numbers from Cura.

Answers

$\text{Rs } 207 \cdot 240 \text{ g}$, about Rs 1,104.





Safety first

Printers are hot and full of moving parts



Hot nozzle: about 200 °C. Never touch it, even after printing: it stays hot for minutes.



Hot bed: about 60 °C. Wait until it cools before removing prints.



Moving parts: keep fingers, hair, sleeves and ID cards away while it runs.



Fresh air: print in a ventilated room. PLA is mild, other plastics can smell.



Adults operate it: students design; a trained adult loads filament, starts and removes prints.



Never leave it alone

Check on long prints regularly. Keep the area clear of paper and clutter.





Fix it

Common print problems

Ages 12+

Ages 15+



What you see

Try this

Print won't stick to the bed

level the bed, clean it, add a brim, check first layer height

'Spaghetti' mess

print came loose: same fixes, and slow down the first layer

Corners lift up (warping)

brim, heated bed on, avoid drafts and fans

Thin hairs between parts (stringing)

lower the nozzle temperature a little, dry the filament

Gaps or weak layers

nozzle partly blocked or filament tangled: ask the teacher

Overhang droops

add supports or rotate the model

First layer

80 % of problems start in the first layer. Watch it go down!





Projects for every age

What each group will design



Little Explorers

ages 5 to 7

Teacher-led: stack shapes into a **robot** or **name keyring**.

Teacher exports and prints.

Focus: shapes and colours.

Junior Makers

ages 8 to 11

Name badge, then a measured **sensor mount** for the robot (JM-10). Focus: holes, align, mm.

Young Innovators

ages 12 to 14

Plant monitor enclosure with cable holes (YI-16). Focus: tolerances, slicing, cost.

Tech Leaders

ages 15 to 17

Parametric robot gripper in Onshape or Fusion (TL-15). Focus: parameters, strength, iteration.



Same rule for everyone

Design → print → **test on the real thing** → improve. Version 2 is always better than version 1!

Showcase

Display version 1 next to the final version. Visitors love seeing the improvement!



Class game

Design sprint!

How to play

1. Teams get a real problem card: e.g. **'My pencils keep rolling off the desk.'**
2. 5 minutes: sketch 3 different ideas on paper.
3. Choose one and model it in Tinkercad (20 minutes).
4. Present: what problem, what design, why it will work.
5. Class votes on which designs to print first.

Then discuss

Which design solved the problem in the simplest way? Simple usually wins!

Big idea

Good design starts with a real problem and a real user, not with the printer.





Make responsibly

Think before you print



Waste less: check the design and the Cura preview before printing. Every failed print is plastic waste.



Credit others: remixing a design from a sharing site? Follow its licence and name the designer.



Never print: weapons or parts that copy real ones, or anything meant to harm.



Respect brands: don't copy logos or famous characters for selling.



Ask: every print in our lab is approved by the teacher first.



Recycle

Keep failed PLA prints in a box. Some companies can recycle PLA into new filament.



Level up!

Level-up challenges

Choose your challenge



Everyone: design a keyring with your initials that is under 10 g of plastic.



Ages 8+: design a box with a lid that fits (use a 0.4 mm gap).



Ages 12+: design two parts that click together with a snap-fit hook.



Ages 15+: use Tinkercad Codeblocks or OpenSCAD to generate a design from code.

Think like an engineer

Why do engineers usually print a cheap, fast version first instead of the perfect final part?

Boss level

Print a 'draft' at 0.3 mm layers and 10 % infill to test the fit in half the time.



Show what you know

What did we learn?

- ☒ 3D printers add plastic layer by layer (additive)
- ☒ Tinkercad: solids + holes, grouped, measured in mm
- ☒ Design rules: flat base, 45° overhangs, thick walls
- ☒ Leave a tolerance gap so parts fit
- ☒ Slice in Cura, print safely, test, improve

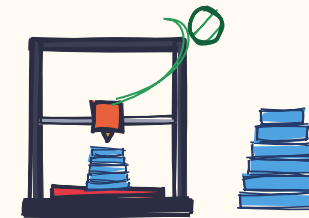


Exit ticket: why won't a 10 mm peg fit a 10 mm printed hole?



At home

Find 3 plastic objects at home. Could any be 3D printed?
Which side would you print it on?



Next class: your group's 3D design class!

